

Statistical Machine Learning (BE4M33SSU) Probably Approximately Correct Learning

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V. Franc

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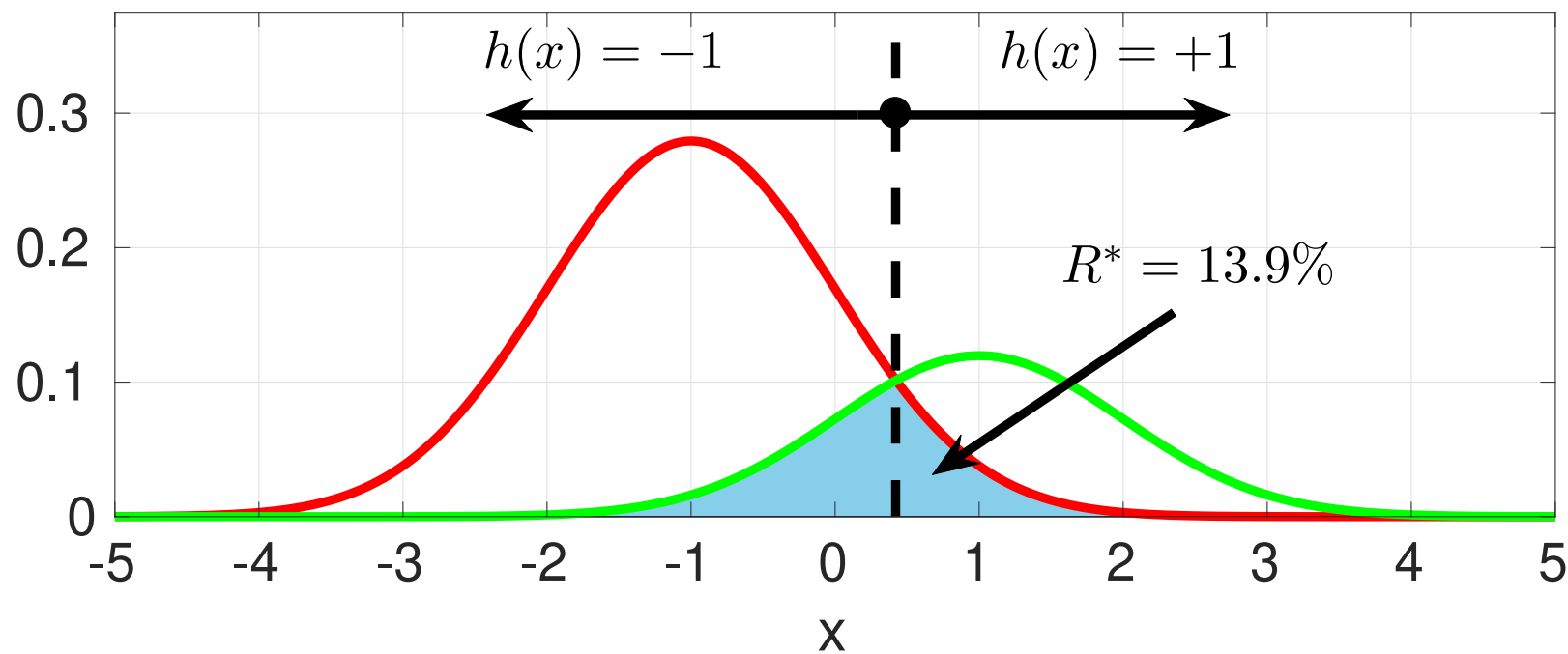
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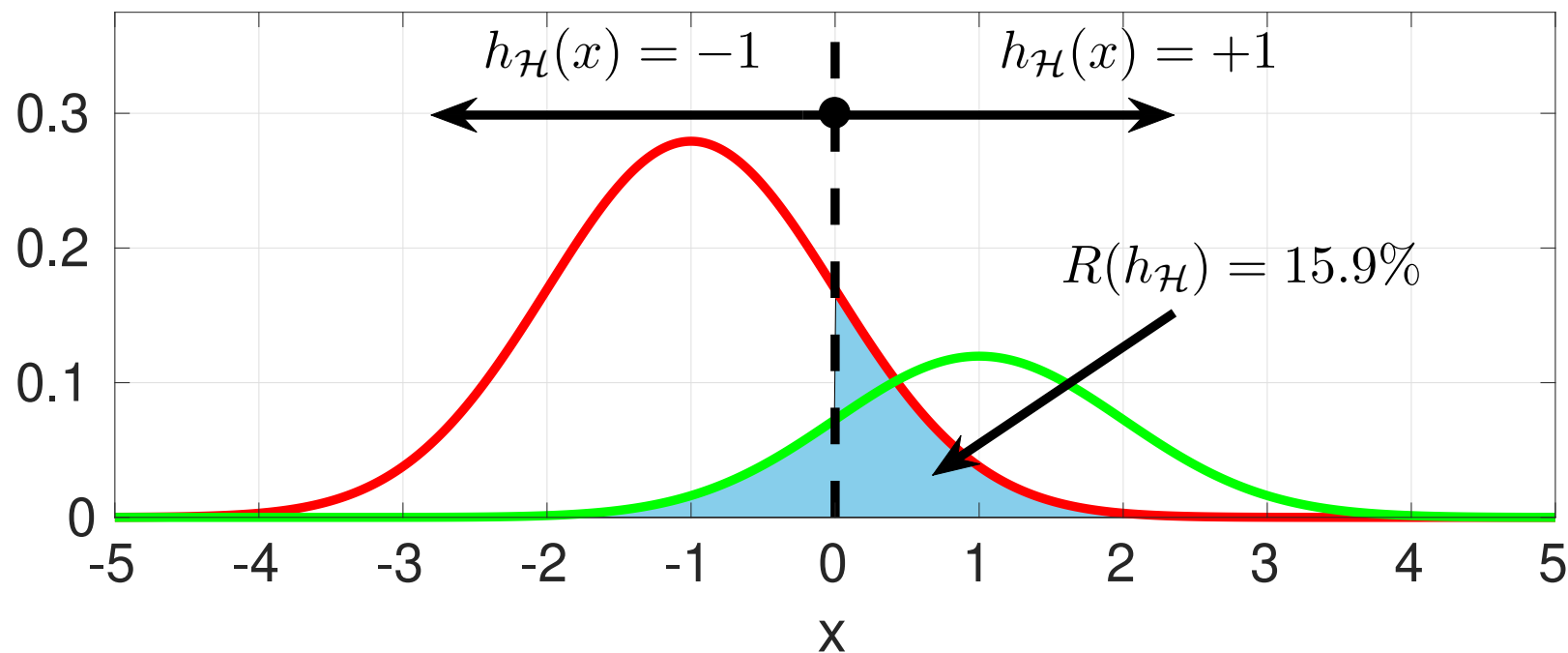


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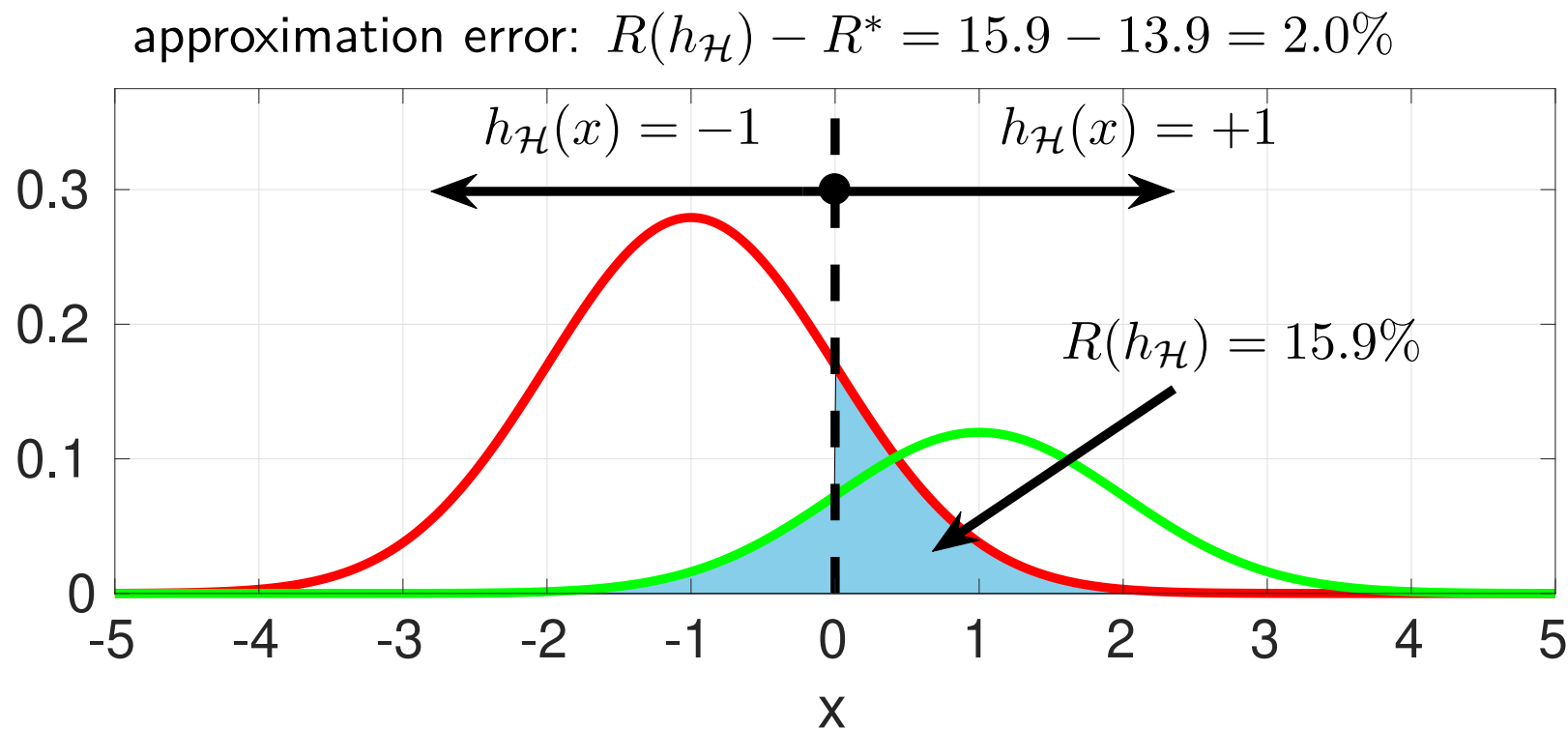
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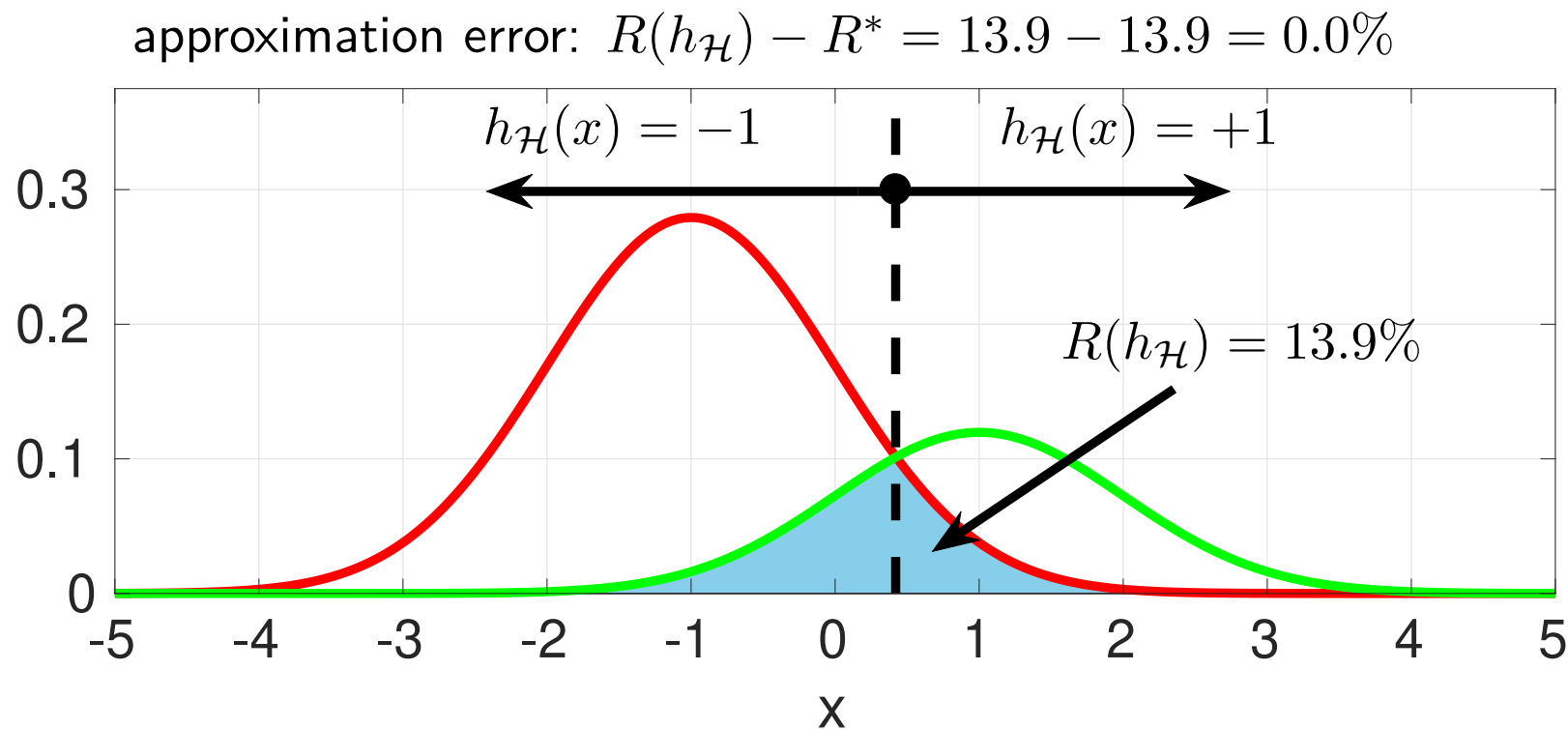
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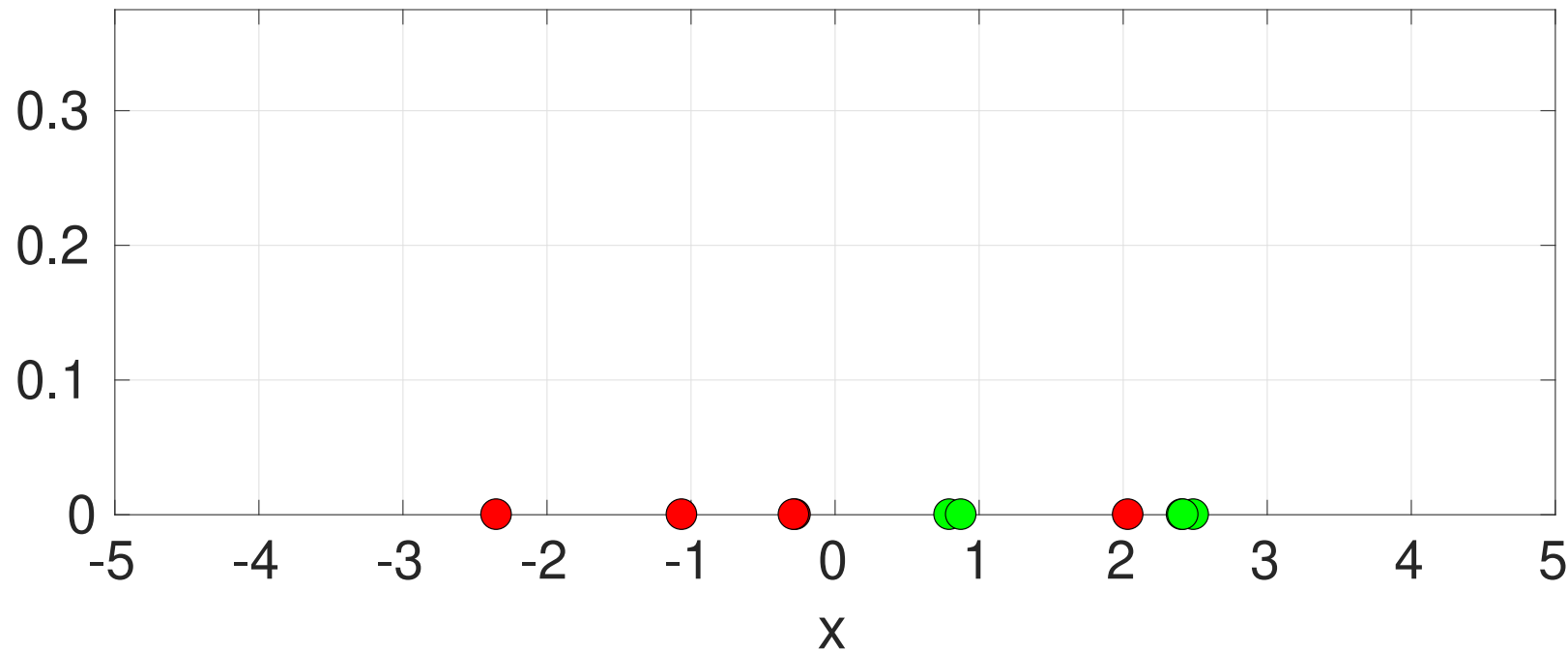
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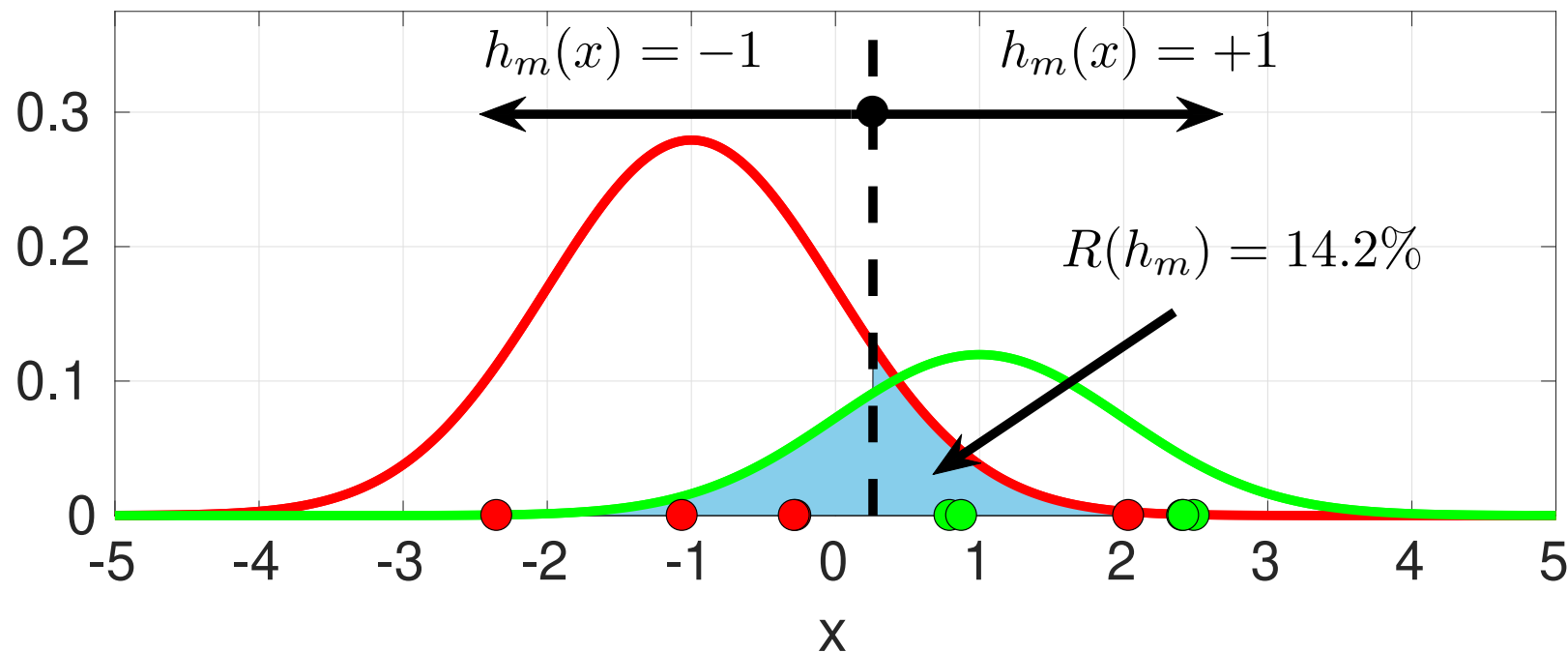
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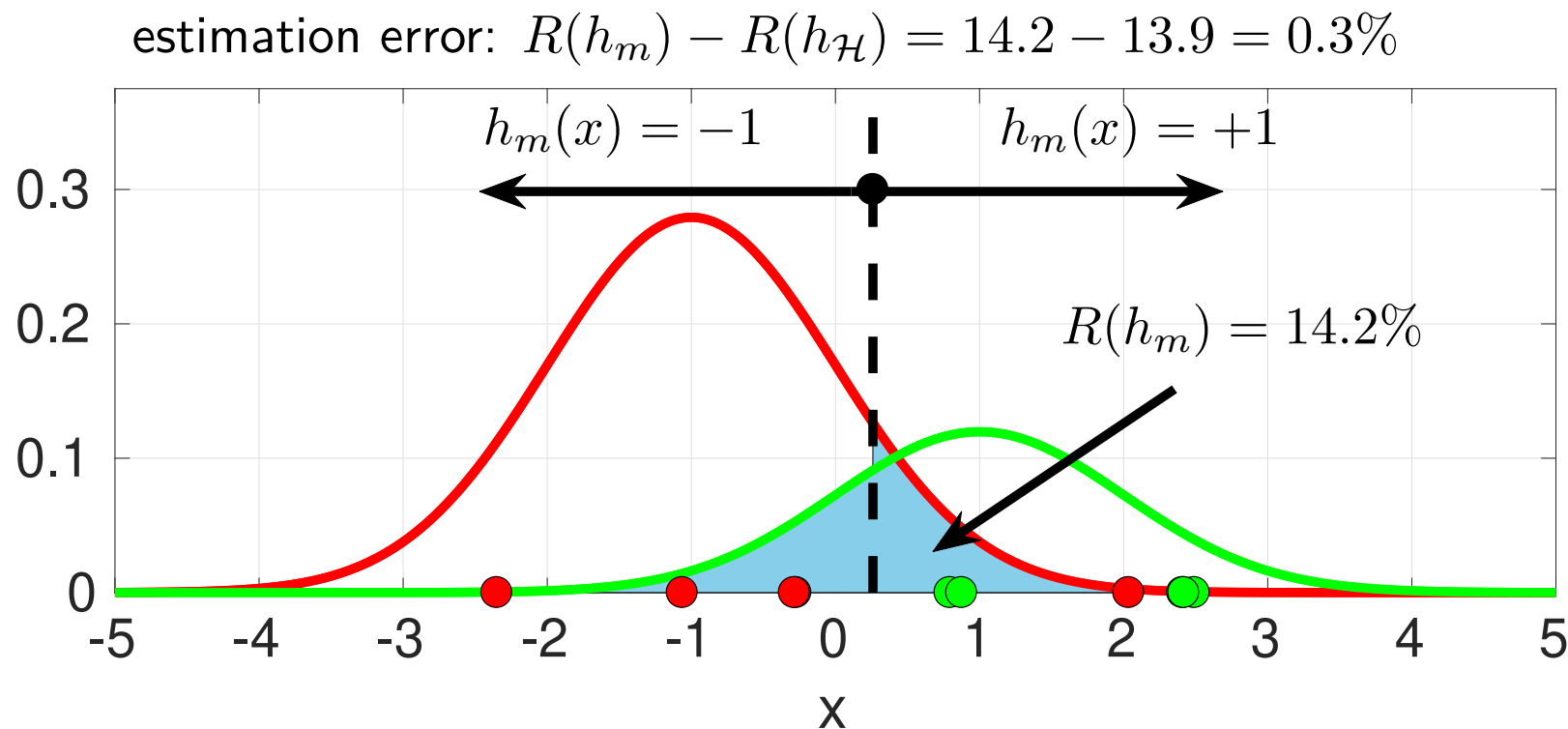
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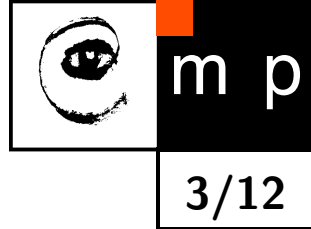
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- ◆ R^* is the irreducible error.
- ◆ The approximation error: depends on $\mathcal{H}$ chosen prior to learning.
- ◆ The estimation error: depends on $\mathcal{H}$, data $\mathcal{T}$ and the algorithm A .

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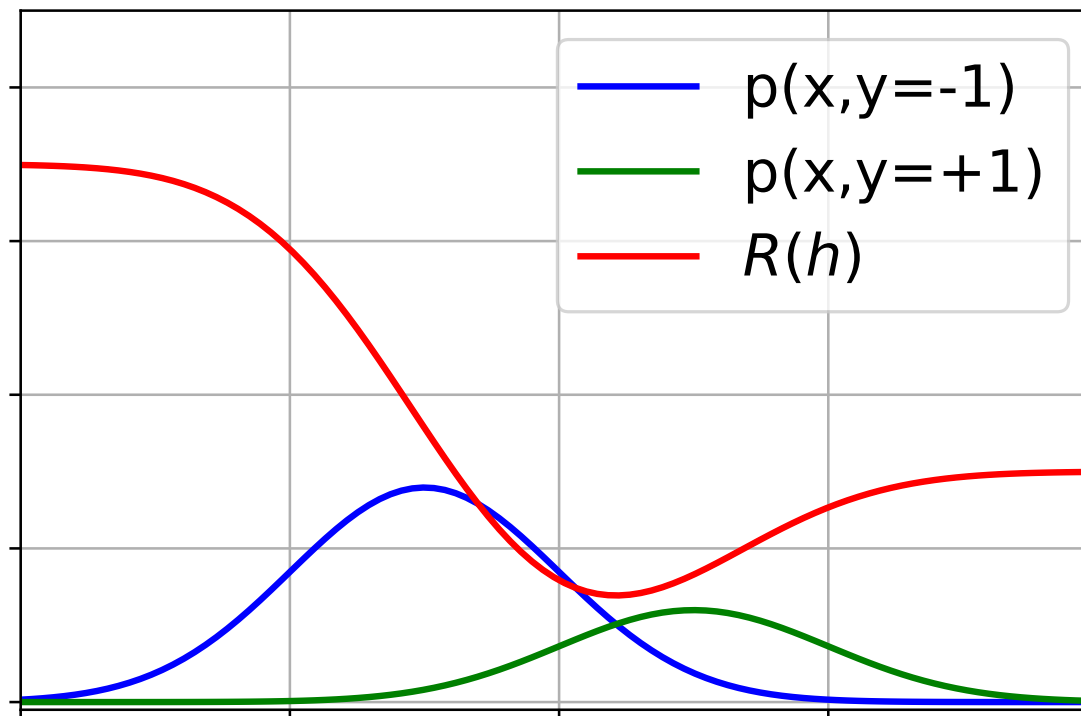
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- ◆ Remarks:
 - The guarantee has to be valid for any data distribution $p(x, y)$.
 - The function $m_{\mathcal{H}}^{\text{pac}}(\varepsilon, \delta)$ is called the sample complexity.
 - The meaning of approximately correct is specified by $\varepsilon \in \mathbb{R}_{>0}$.
 - The meaning of probably correct is specified by $\delta \in (0, 1)$.

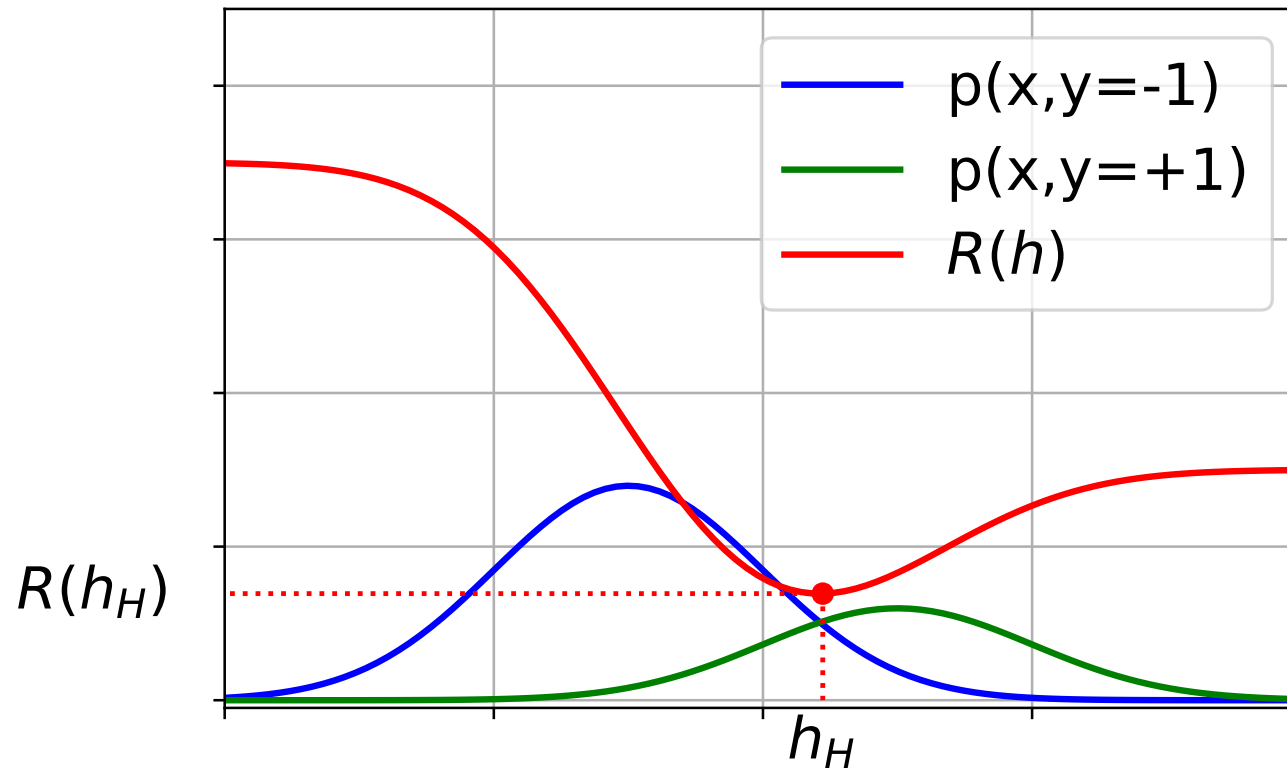
ULLN implies that ERM is successful PAC learner

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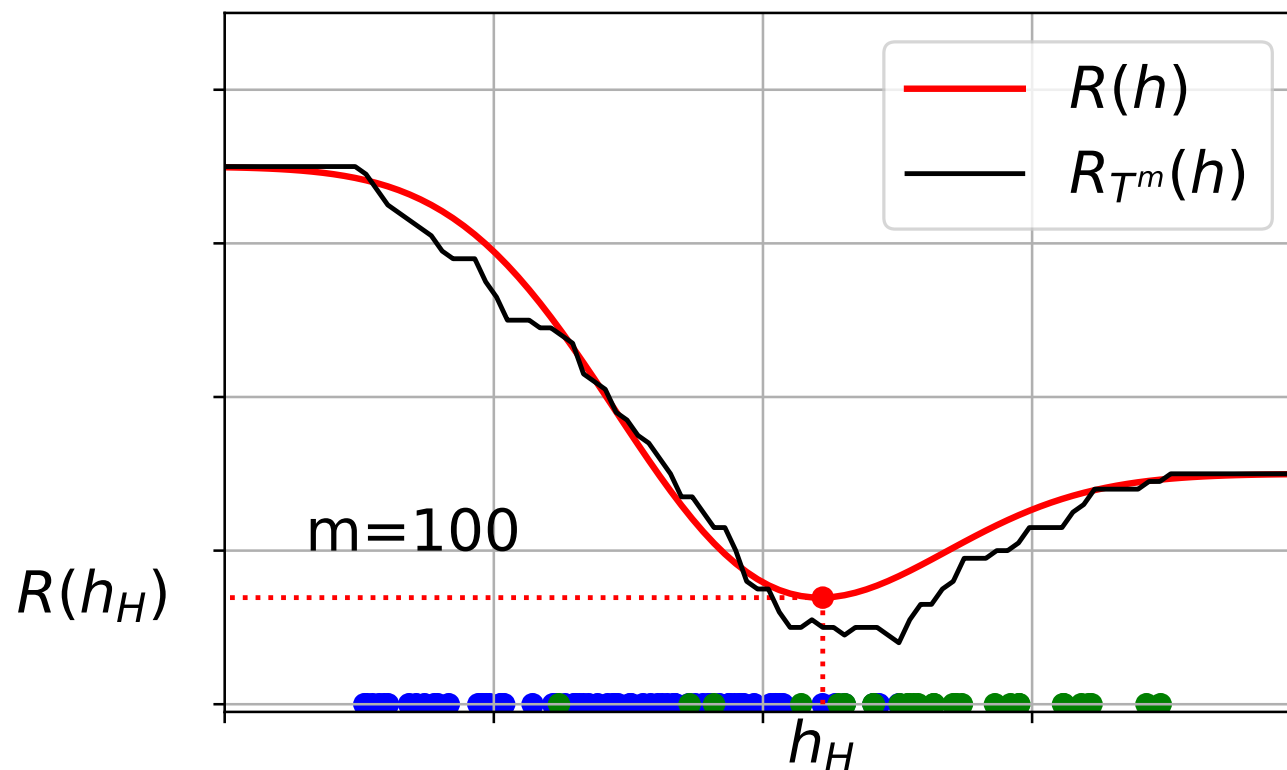
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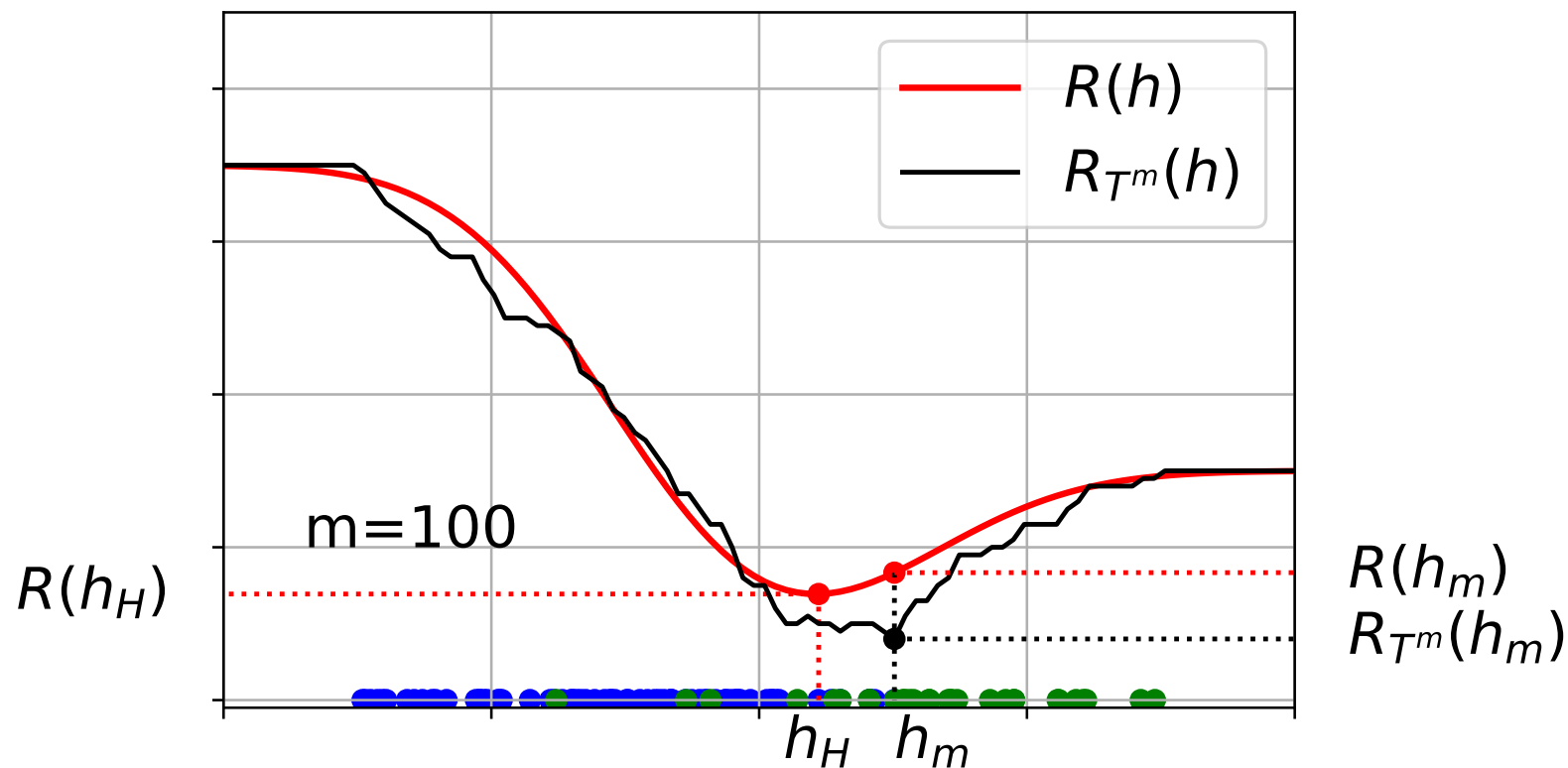
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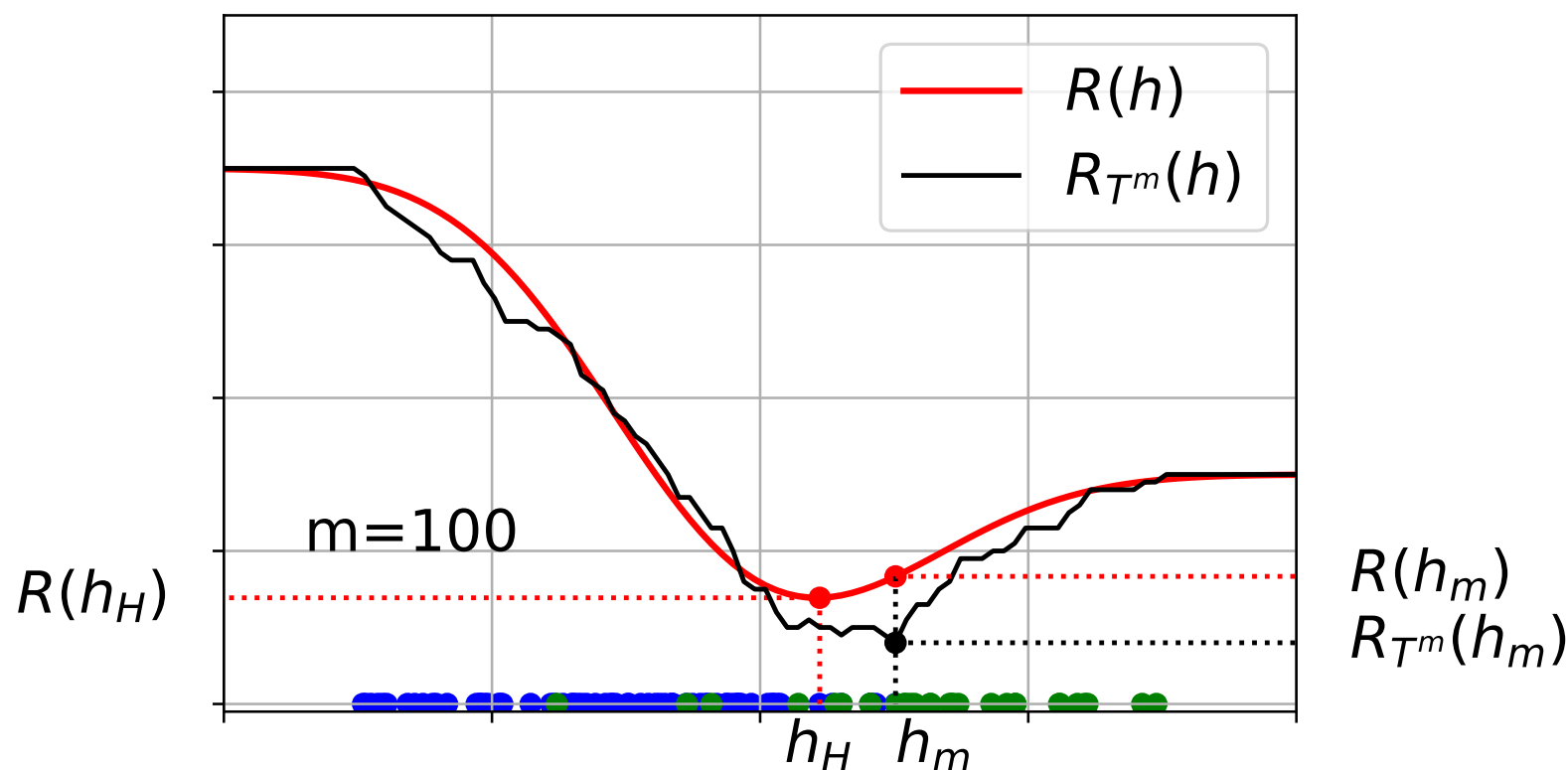
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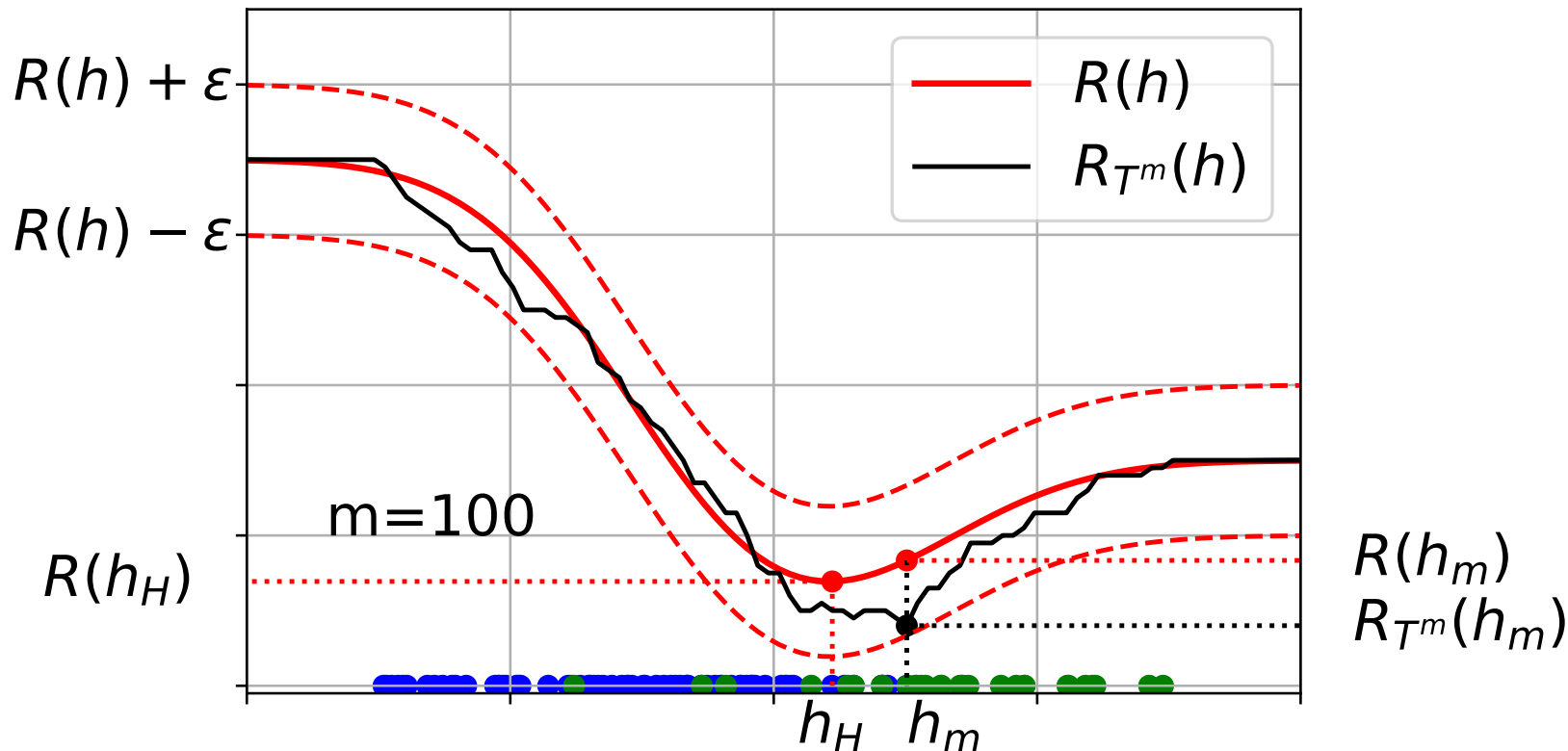
ULLN:

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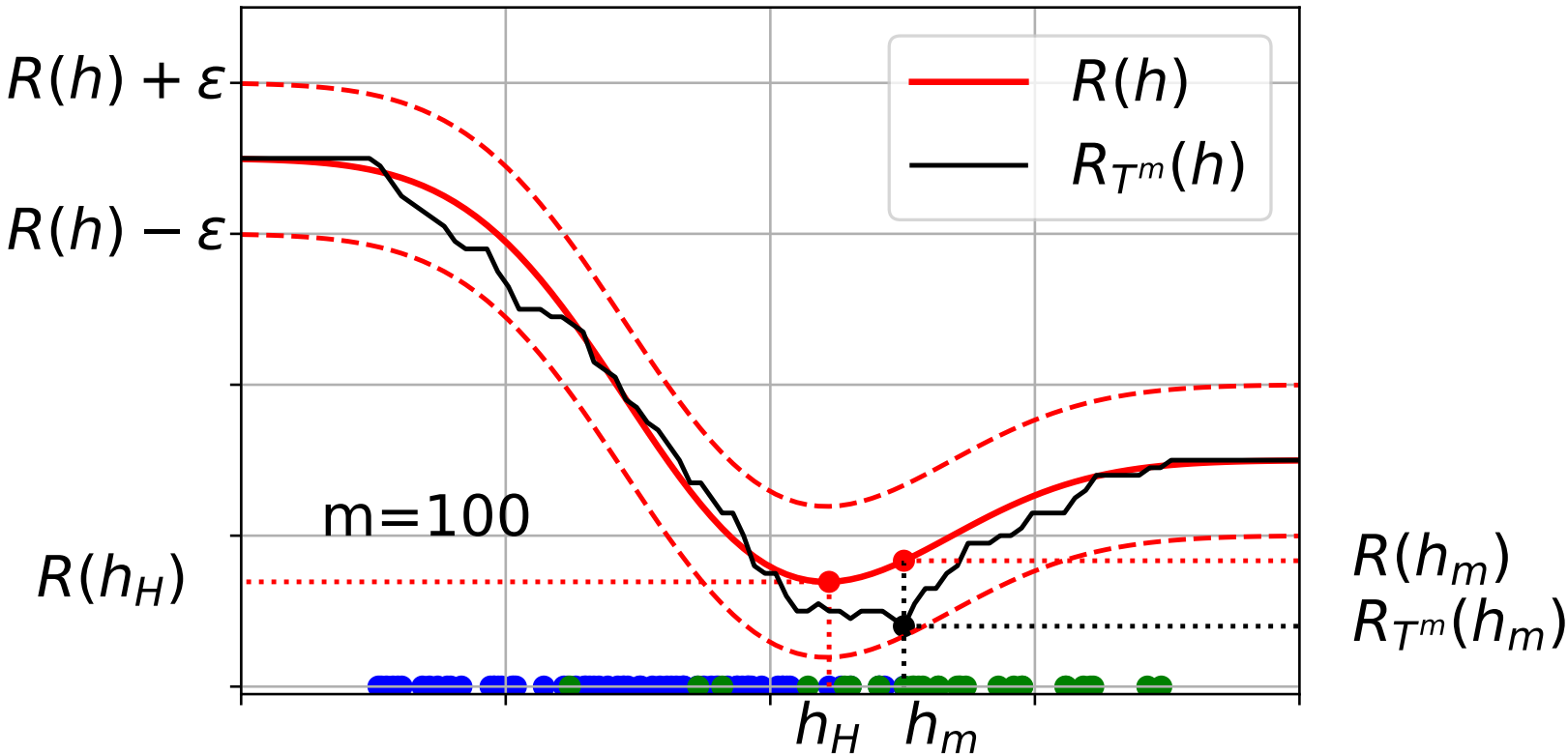
$$\begin{aligned} \underbrace{R(h_m) - R(h_{\mathcal{H}})}_{\text{estimation error}} &= \left(R(h_m) - R_{\mathcal{T}^m}(h_m)\right) + \left(R_{\mathcal{T}^m}(h_m) - R(h_{\mathcal{H}})\right) \\ &\leq \left(R(h_m) - R_{\mathcal{T}^m}(h_m)\right) + \left(R_{\mathcal{T}^m}(h_{\mathcal{H}}) - R(h_{\mathcal{H}})\right) \\ &\leq 2 \sup_{h \in \mathcal{H}} |R(h) - R_{\mathcal{T}^m}(h)| \end{aligned}$$

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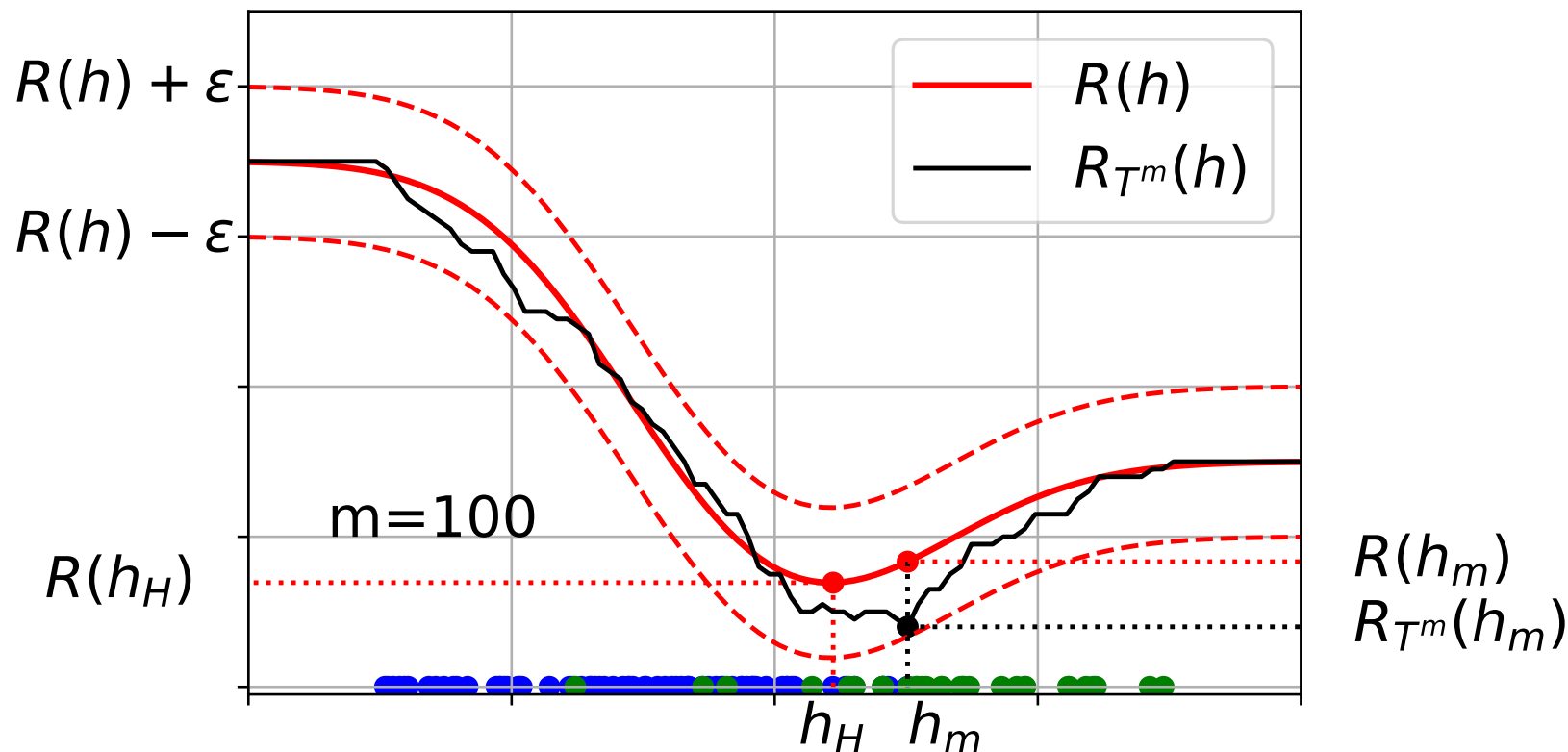
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- ◆ Successful PAC learner for $\mathcal{H} \subset \mathcal{Y}^{\mathcal{X}}$: there exists $m_{\mathcal{H}}^{\text{pac}}: \mathbb{R}_{>0} \times (0, 1) \rightarrow \mathbb{N}$ such that when running the algorithm on $m \geq m_{\mathcal{H}}^{\text{pac}}(\bar{\varepsilon}, \delta)$ examples $\mathcal{T}^m \sim p^m$ then it returns $h_m = A(\mathcal{T}^m)$ such that

$$\mathbb{P}\left(\underbrace{R(h_m) - R(h_{\mathcal{H}})}_{\text{approximately correct}} \leq \bar{\varepsilon}\right) \geq 1 - \delta.$$

- ◆ **Theorem:** If ULLN applies for $\mathcal{H} \subset \mathcal{Y}^{\mathcal{X}}$ then ERM $h_m \in \text{Argmin}_{h \in \mathcal{H}} R_{\mathcal{T}^m}(h)$ is a successful PAC learner for $\mathcal{H}$ with the sample complexity $m_{\mathcal{H}}^{\text{pac}}(\bar{\varepsilon}, \delta) = m_{\mathcal{H}}^{\text{ul}}(\frac{\bar{\varepsilon}}{2}, \delta)$.

ERM is successful PAC learner for finite hypothesis space

- ◆ ULLN applies for finite hypothesis space $\mathcal{H} = \{h_1, \dots, h_K\}$:

$$\mathbb{P}\left(\max_{h \in \mathcal{H}} |R_{\mathcal{T}^m}(h) - R(h)| \geq \varepsilon\right) \leq 2|\mathcal{H}|e^{-\frac{2m\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2}} = \delta$$

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- ◆ **Corollary:** ERM is successful PAC learner for $\mathcal{H}$ with sample complexity

$$m_{\mathcal{H}}^{\text{pac}}(\bar{\varepsilon}, \delta) = 2\frac{\log 2|\mathcal{H}| - \log \delta}{\bar{\varepsilon}^2}(\ell_{\max} - \ell_{\min})^2,$$

that is, when running ERM on $\mathcal{T}^m$ with $m \geq m_{\mathcal{H}}^{\text{pac}}(\bar{\varepsilon}, \delta)$ then it returns $h_m = A(\mathcal{T}^m)$ such that

$$\mathbb{P}\left(R(h_m) - R(h_{\mathcal{H}}) \leq \bar{\varepsilon}\right) \geq 1 - \delta.$$

Linear classifier minimizing classification error

- ◆ $\mathcal{X}$ is a set of observations and $\mathcal{Y} = \{+1, -1\}$ a set of hidden labels
- ◆ $\phi: \mathcal{X} \rightarrow \mathbb{R}^n$ is fixed feature map embedding $\mathcal{X}$ to $\mathbb{R}^n$
- ◆ **Task:** find linear classification strategy $h: \mathcal{X} \rightarrow \mathcal{Y}$, parametrized by a vector $\mathbf{w} \in \mathbb{R}^n$,

$$h(x; \mathbf{w}, b) = \text{sign}(\langle \mathbf{w}, \phi(x) \rangle + b) = \begin{cases} +1 & \text{if } \langle \mathbf{w}, \phi(x) \rangle + b \geq 0 \\ -1 & \text{if } \langle \mathbf{w}, \phi(x) \rangle + b < 0 \end{cases}$$

with minimal expected risk

$$R^{0/1}(h) = \mathbb{E}_{(x,y) \sim p} \left(\ell^{0/1}(y, h(x)) \right) \quad \text{where} \quad \ell^{0/1}(y, y') = \mathbb{I}[y \neq y']$$

- ◆ We are given a set of training examples

$$\mathcal{T}^m = \{(x^i, y^i) \in (\mathcal{X} \times \mathcal{Y}) \mid i = 1, \dots, m\}$$

drawn from i.i.d. with the distribution $p(x, y)$.

ERM learning for linear classifiers

- ◆ ERM for $\mathcal{H} = \{h(x; \mathbf{w}, b) = \text{sign}(\langle \mathbf{w}, \phi(x) \rangle + b) \mid (\mathbf{w}, b) \in \mathbb{R}^{n+1}\}$ leads to

$$(\mathbf{w}^*, b^*) \in \underset{h \in \mathcal{H}}{\text{Argmin}} R_{\mathcal{T}_m}^{0/1}(h) = \underset{(\mathbf{w}, b) \in (\mathbb{R}^n \times \mathbb{R})}{\text{Argmin}} R_{\mathcal{T}_m}^{0/1}(h(\cdot; \mathbf{w}, b)) \quad (1)$$

where the empirical risk is

$$R_{\mathcal{T}_m}^{0/1}(h(\cdot; \mathbf{w}, b)) = \frac{1}{m} \sum_{i=1}^m [y^i \neq h(x^i; \mathbf{w}, b)]$$

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- Algorithmic issues (lecture on Support Vector Machines): in general, there is no known algorithm solving the task (1) in time polynomial in m .
- Does ULLN applies for the class of two-class linear classifiers?
...If yes we known that ERM is PAC successful learner.

Vapnik-Chervonenkis (VC) dimension

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Definition: Let $\mathcal{H} \subseteq \{-1, +1\}^{\mathcal{X}}$. The **Vapnik-Chervonenkis dimension** of $\mathcal{H}$ is the cardinality of the largest set of points from $\mathcal{X}$ which can be shattered by $\mathcal{H}$.

VC dimension of class of two-class linear classifiers

Theorem: The VC-dimension of the hypothesis class of all two-class linear classifiers operating in n -dimensional feature space

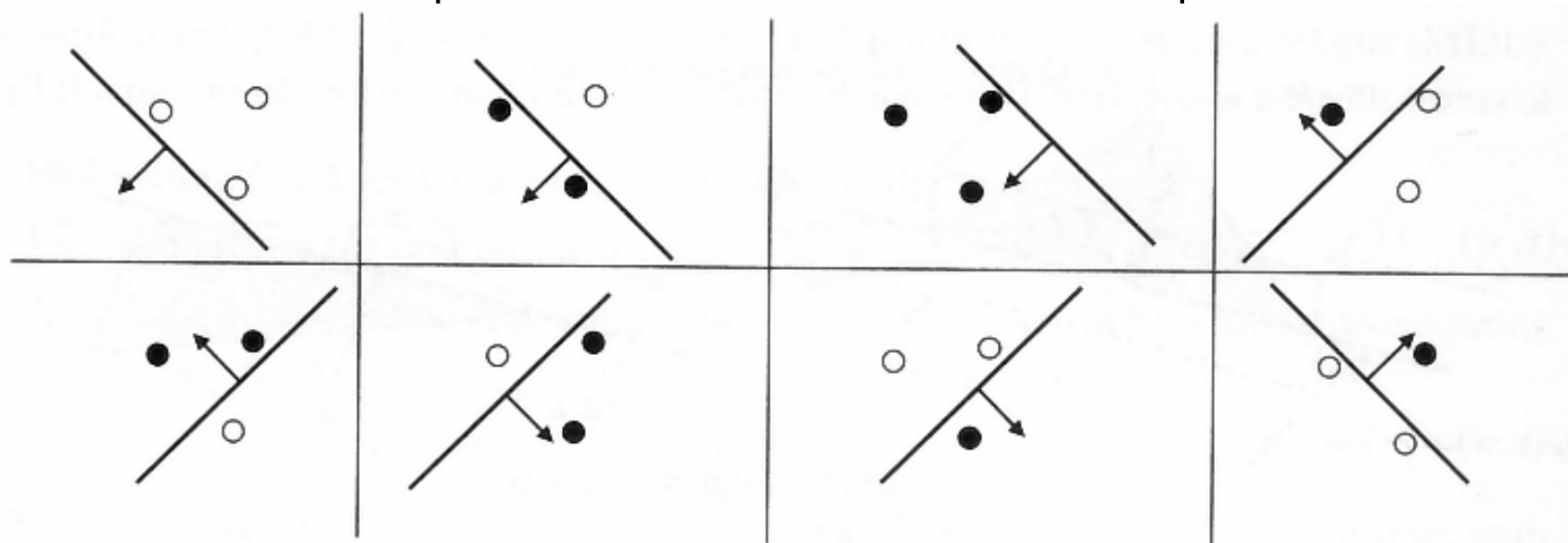
$\mathcal{H} = \{h(x; \mathbf{w}, b) = \text{sign}(\langle \mathbf{w}, \phi(x) \rangle + b) \mid (\mathbf{w}, b) \in (\mathbb{R}^n \times \mathbb{R})\}$ is $n + 1$.

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Example for $n = 2$ -dimensional feature space



ULLN for two class predictors and 0/1-loss

Theorem: Let $\mathcal{H} \subset \{+1, -1\}^{\mathcal{X}}$ be a hypothesis class with VC dimension $d < \infty$ and $\mathcal{T}^m = \{(x^1, y^1), \dots, (x^m, y^m)\} \in (\mathcal{X} \times \mathcal{Y})^m$ a training set draw from i.i.d. rand vars with distribution $p(x, y)$. Then for any $\varepsilon > 0$ it holds

$$\mathbb{P}\left(\sup_{h \in \mathcal{H}} \left| R^{0/1}(h) - R_{\mathcal{T}^m}^{0/1}(h) \right| \geq \varepsilon \right) \leq 4 \left(\frac{2em}{d} \right)^d e^{-\frac{m\varepsilon^2}{8}}$$

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Corollary: Let $\mathcal{H} \subset \{+1, -1\}^{\mathcal{X}}$ be a hypothesis class with a **finite VC dimension** $d < \infty$. Then, ULLN applies for $\mathcal{H}$ and there exists a constant C such that

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that is, **ERM is PAC successful learner**.

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Remark: Recall that in case of finite hypothesis space $\mathcal{H} = \{h_1, \dots, h_K\}$ and 0/1-loss we have the sample complexity $m_{\mathcal{H}}^{\text{ul}}(\varepsilon, \delta) = \frac{\log 2|\mathcal{H}| - \log \delta}{\varepsilon^2}$.

Summary

- ◆ Error decomposition:
Generalization error = estimation error + approximation error + Bayes risk.
- ◆ Probably Approximately Correct (PAC) learning.
- ◆ ULLN implies that ERM is successful PAC learner.
- ◆ VC dimension: hypothesis space complexity of two-class classifier.
- ◆ VC dimension of linear hypothesis space.
- ◆ Finite VC dimension implies that ERM is a successful PAC learner.

